

Functions in JS

Block of code that performs a specific task, can be invoked whenever needed



means can be called

used to remove redundancy,
redundancy- repeat

Functions in JS

Function Definition

```
function functionName( ) {  
  
    //do some work  
  
}
```

```
function functionName( param1, param2 ...) {  
  
    //do some work  
  
}
```

Function Call

```
functionName( );
```

in the modern day programming of js we use arrow functions only

Arrow Functions

Compact way of writing a function

```
const functionName = ( param1, param2 ...) => {  
  
    //do some work  
  
}
```

```
const sum = ( a, b ) => {  
  
    return a + b;  
  
}
```

arrow fn is compact way of writing fn

it starts directly with paranthesis followed by an arrow => and then block of code which is to be done

like

```
(a,b)=>{  
    console.log(a+b);  
}
```

but if u run it wont show anything so u need to save it in a variable

like

```
const sum=(a,b)=>{  
    console.log(a+b);  
}
```

and see from line 50;

Let's Practice

Qs. Create a function using the “function” keyword that takes a String as an argument & returns the number of vowels in the string.

Qs. Create an arrow function to perform the same task.

Apna College

forEach Loop in Arrays

its a method because associated w arrays.

arr.forEach(callbackFunction) ← the whole definition of the callback fn to be written see line 133

CallbackFunction : Here, it is a function to execute for each element in the array

***A callback is a function passed as an argument to another function.**

callback fn is a fn which is passed as an argument to another fn

```
arr.forEach( ( val ) => {  
    console.log(val);  
} )
```

1. it is only for arrays and not strings
2. in js u can pass fns as arguments and also can return fn value, like fn is a function so u can do return fn;
3. eg line 124

forEach is a higher order function

****IMP interview **what are higher order functions?HOF or HOM*****
--> those fn which either takes another fn as their parameter OR returns a another function
those functions are HOF or HOM

basically foreach loop is just another way to execute the code to each element of the array, you can also do this by the for of loop or the .. normal for loop and run an iterator there... like `for(let i=0;i<arr.length;i++){}` like this but foreach loop has smth special, yes to run the code for all elements you can use any of the loops but the foreach loops can also print the index or can say can directly access the idx so you can print the exact idx

Number:
`arr.forEach(function printval(val,idx,arr){
 console.log(val);
});`

the syntax for foreach loop is

```
arr.forEach(function fn(val,idx,arr){  
  console.log(val,idx,arr);  
});
```

here also write val,idx,arr not only in callback fns argument but here also

in the function definition

`function fn(value,index,array)`

this is the syntax first place is of value whatever you name the value as then after comma write 'idx' or anything that'll be the index then after comma write the array's name to print the array if wanted to and also idx only if wanted to

Let's Practice

Qs. For a given array of numbers, print the square of each value using the forEach loop.

mostly inside the forEach method , the callback fn is in the form of arrow fn like this

```
arr.forEach((val)=>{  
  block of code}  
);
```

ONE THING TO REMEMBER IS THAT U DONT
NEED TO CALL THE FUNCTION AGAIN IT
AUTOMATICALLY EXECUTES THE FN ...
IN EITHER ARROW FN OR NORMAL FN
BUT FOREACH KE ANDAR

Some More **Array Methods**

Map

Creates a new array with the results of some operation. The value its callback returns are used to form new array

`arr.map( callbackFnx( value, index, array ) )`

```
let newArr = arr.map( ( val ) => {  
    return val * 2;  
} )
```

the returned array should be stored it doesnt automatically stores so store it in a new let arr array variable

and to make a new arr u need to return some values from the map method , those values will eventually get stored in the new arr only then itll get saved into the new array variable

map is just like the foreach loop which does operation on each element of the array but the only difference is map method returns you a new array and foreach doesnt create a new array and not just prints the operated array.

the syntax is like the forEach method only

Some More **Array Methods**

Filter

Creates a new array of elements that give true for a condition/filter.

Eg: all even elements

```
let newArr = arr.filter( ( val ) => {  
    return val % 2 === 0;  
})
```

filter is not normal like other method we can also do this ques from map method but in that we need to do `if(val%2==0)return val;` which is normal but filter method me u just need to return the condition and itll automatically stores the value if tis true like see the example in this page only

filter method means it filters some elements and stores only those elements which satisfies some condition.. like example from a given array store only the even elements in the new one so u use filter method and give out a condition.

syntax is just like foreach and map method , but here u return some operated elements like map method cuz u want to create a new array

like sum, average etc

Some More **Array Methods**

Reduce

Performs some operations & reduces the array to a single value. It returns that single value.

see code line 184

JavaScript Demo: Array.reduce()

```
1 const array1 = [1, 2, 3, 4];
2
3 // 0 + 1 + 2 + 3 + 4
4 const initialValue = 0;
5 const sumWithInitial = array1.reduce(
6   (accumulator, currentValue) => accumulator + currentValue,
7   initialValue,
8 );
9
10 console.log(sumWithInitial);
11 // Expected output: 10
```

syntax is like foreach and map and filter method only here also u return and use callback fn

Let's Practice

Qs. We are given array of marks of students. Filter out of the marks of students that scored 90+.

Qs. Take a number n as input from user. Create an array of numbers from 1 to n.

- Use the reduce method to calculate sum of all numbers in the array.**
- Use the reduce method to calculate product of all numbers in the array.**